

Task 6 – Introduction to Cryptography

Tools Used

- Ubuntu Linux
- OpenSSL

1) Cryptography.

Cryptography is the method of protecting information so that only authorized users can read it. It ensures:

- **Confidentiality** – data remains secret
- **Integrity** – data is not modified
- **Authentication** – identity is verified
- **Non-repudiation** – sender cannot deny sending

2) Symmetric vs Asymmetric Encryption.

Symmetric encryption uses the same key to encrypt and decrypt data. AES (Advanced Encryption Standard) is a popular symmetric encryption algorithm and is fast and secure. It is commonly used for encrypting files and network traffic.

Asymmetric encryption uses two keys: a public key and a private key. RSA is the most commonly used asymmetric encryption method. The public key is shared, and the private key is kept secret. It is mainly used for secure communication and digital signatures.

Type	Key Usage	Example	Usage
Symmetric	Same key for encrypt & decrypt	AES	File encryption, VPN
Asymmetric	Public + Private key	RSA	HTTPS, Email, digital signature

3) AES File Encryption.

Step 1 – Create sample file

```
mounika@ubuntu: ~  
mounika@ubuntu:~$ nano test.txt  
mounika@ubuntu:~$ cat test.txt  
This is my Task6.  
And i am Mounika.  
mounika@ubuntu:~$
```

Step 2 – Encrypt using AES

```
mounika@ubuntu:~$ openssl enc -aes-256-cbc -salt -in test.txt -out test_encrypte  
d.txt  
enter AES-256-CBC encryption password:  
Verifying - enter AES-256-CBC encryption password:  
*** WARNING : deprecated key derivation used.  
Using -iter or -pbkdf2 would be better.  
mounika@ubuntu:~$ cat test_encrypted.txt  
Salted__  
mounika@ubuntu:~$ cat test.txt  
This is my Task6.  
And i am Mounika.  
mounika@ubuntu:~$ cat test_encrypted.txt  
Salted__
```

Step 3 – Decrypt the file

```

mounika@ubuntu:~$ openssl enc -aes-256-cbc -d -in test_encrypted.txt -out test_decrypt.txt
enter AES-256-CBC decryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
mounika@ubuntu:~$ cat test.txt
This is my Task6.
And i am Mounika.
mounika@ubuntu:~$ cat test_encrypted.txt
Salted__U&p)M_W5@aXN-J-rA!fOpZ<k
mounika@ubuntu:~$ cat test_decrypt.txt
This is my Task6.
And i am Mounika.
mounika@ubuntu:~$

```

4) RSA Key Generation.

Step 1 – Generate private key

```
openssl genrsa -out private.pem 2048
```

Step 2 – Generate public key

```
openssl rsa -in private.pem -pubout -out public.pem
```

```

mounika@ubuntu:~$ openssl genrsa -out private.pem 2048
mounika@ubuntu:~$ openssl rsa -in private.pem -pubout -out public.pem
writing RSA key
mounika@ubuntu:~$

```

```

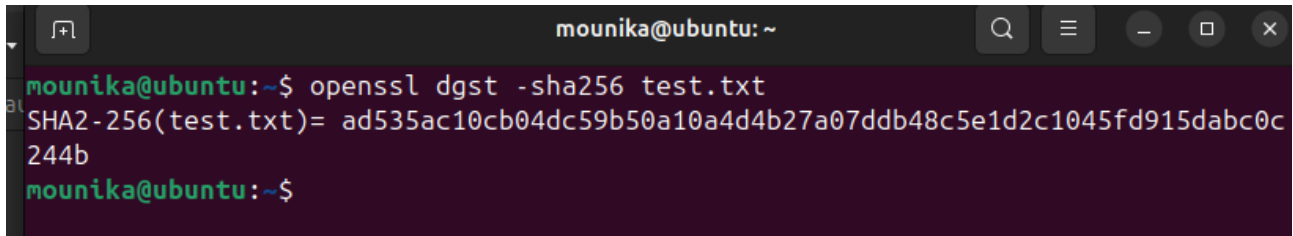
private.pem
Public
public.pem

```

5) Hashing & Integrity.

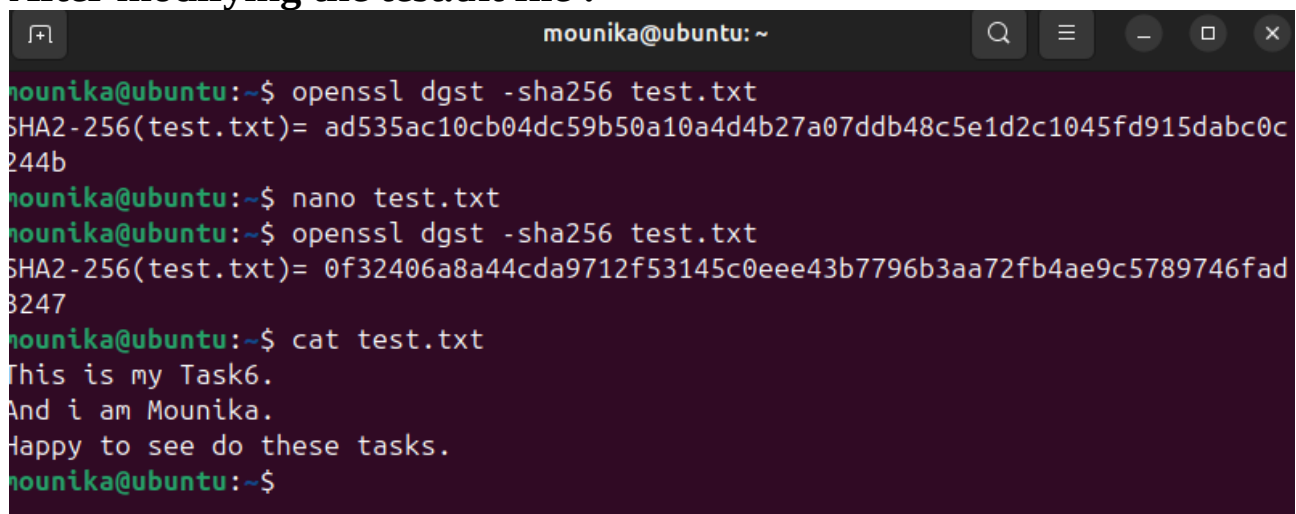
Step 1 – Generate hash

`openssl dgst -sha256 test.txt`



```
mounika@ubuntu: ~  
mounika@ubuntu:~$ openssl dgst -sha256 test.txt  
SHA2-256(test.txt)= ad535ac10cb04dc59b50a10a4d4b27a07ddb48c5e1d2c1045fd915dabc0c  
244b  
mounika@ubuntu:~$
```

After modifying the test.txt file :



```
mounika@ubuntu:~$ openssl dgst -sha256 test.txt  
SHA2-256(test.txt)= ad535ac10cb04dc59b50a10a4d4b27a07ddb48c5e1d2c1045fd915dabc0c  
244b  
mounika@ubuntu:~$ nano test.txt  
mounika@ubuntu:~$ openssl dgst -sha256 test.txt  
SHA2-256(test.txt)= 0f32406a8a44cda9712f53145c0eee43b7796b3aa72fb4ae9c5789746fad  
8247  
mounika@ubuntu:~$ cat test.txt  
This is my Task6.  
And i am Mounika.  
Happy to see do these tasks.  
mounika@ubuntu:~$
```

6) Real-World Use of Cryptography.

1.HTTPS – Secure Websites

- HTTPS is used in websites like <https://gmail.com>
- The “S” in HTTPS stands for **Secure**
- It protects:
 - Passwords
 - Messages
 - Personal information

How HTTPS works using cryptography:

1. Gmail sends its **RSA public key** to the browser.

2. The browser creates a **secret AES key**.
3. The AES key is encrypted using Gmail's **RSA public key**.
4. Gmail decrypts the AES key using its **private key**.
5. Both Gmail and the browser use **AES** to encrypt all data.
6. **Hashing** is used to make sure data is not changed.

This ensures:

- Data confidentiality
- Secure login
- Protection from hackers

2.VPN – Secure Internet Tunnel

- VPN stands for **Virtual Private Network**
- It creates a **secure encrypted tunnel** between the device and the internet
- It is very useful on **public Wi-Fi** like in cafes or airports

How VPN uses cryptography:

1. The device and VPN server exchange keys using **RSA**.
2. A **secret AES key** is created.
3. All internet traffic is encrypted using **AES**.
4. **Hashing** checks that the data is not modified.
5. Hackers can only see **scrambled data**.

This provides:

- Privacy
- Security
- Protection from spying

7) Algorithm Comparison.

Algorithm	Type	Purpose
AES	Symmetric	File & data encryption
RSA	Asymmetric	Secure key exchange
SHA-256	Hash	Data integrity
MD5	Hash	Weak, not secure

8) Conclusion.

From this task, I learned how cryptography protects digital information. AES was used to encrypt files securely, RSA keys were generated for secure communication, and hashing was used to verify data integrity. I also learned how HTTPS uses all these technologies together to protect websites like Gmail and online banking. This experiment helped me understand how cryptography works in real-world security systems.